

A Second-Order Implicit-Explicit Scheme for the Baroclinic-Barotropic Split System of Primitive Equations

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Abstract. The baroclinic-barotropic mode splitting technique is commonly employed in numerical solutions of the primitive equations for ocean modeling to deal with the multiple time scales of ocean dynamics. In this paper, a second-order implicit-explicit (IMEX) scheme is proposed to advance the baroclinic-barotropic split system. Specifically, the baroclinic mode and the layer thickness of fluid are evolved explicitly via the second-order strong stability preserving Runge-Kutta scheme, while the barotropic mode is advanced implicitly using the linearized Crank-Nicolson scheme. At each time step, the baroclinic velocity is first computed using an intermediate barotropic velocity. The barotropic velocity is then corrected by re-advancing the barotropic mode with an improved barotropic forcing. Finally, the layer thickness is updated by coupling the baroclinic and barotropic velocities together. In addition, numerical inconsistencies on the discretized sea surface height caused by the mode splitting are alleviated via a reconciliation process with carefully calculated flux deficits. Temporal truncation error is also analyzed to validate the second-order accuracy of the scheme. Finally, two benchmark tests from the MPAS-Ocean platform are conducted to numerically demonstrate the performance of the proposed IMEX scheme.

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Key words: Primitive equations, baroclinic-barotropic splitting, implicit-explicit, strong stability preserving RK, SSH reconciliation.

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1 Introduction

The ocean is a critical component of the climate and plays a significant role in our daily lives. To comprehend and simulate ocean dynamics, various computational models have been developed based on the fundamental laws of physics and the characteristics of geophysical flows. The primitive equations [1–3], a simplification of the Navier-Stokes equations, are one of these models. They have been widely used in real-world applications and coupled with tracers such as temperature, salinity, and chemicals to fluid velocity, fluid depth, and pressure. Due to the stratification effect of the ocean [4], the fluid is often modeled as a stack of immiscible layers, each with a uniform fluid density. Such layered models simplify the computational costs of stratified fluid flows and perform well at portraying vertical profiles. For additional details, readers can refer to [3, 4] and the references cited therein. Nonetheless, conducting long-term numerical simulations for layered models is still quite challenging because of their extensive computational complexity, which arises from the integration of the large-scale oceanic system. Specifically, the oceanic dynamics frequently consist of various time scales resulting from the interplay between external and internal gravity waves and the rotation of the Earth. In layered ocean models, the barotropic mode is considered as the fastest of the entire spectrum of inertial-gravity waves, while the remaining part is referred to as the baroclinic mode. The barotropic mode has a wave speed of up to approximately 200 m/s, while the advection-dominated baroclinic mode is only about 2 m/s [5]. Since the barotropic velocity is about two orders of magnitude faster than the baroclinic one, the barotropic mode highly restricts the time step size under the Courant-Friedrichs-Lewy (CFL) condition if the entire system is advanced via an explicit uniform stepping scheme. Therefore, it is natural to split these two modes in ocean dynamics and evolve them separately over time.

The barotropic-baroclinic mode splitting technique has been extensively studied in the literature [6–14]. This approach splits the primitive equations into two modes: the barotropic mode, which is obtained through vertical averaging and is a 2D mode, and the baroclinic mode, which is the difference between the original velocity and the barotropic one and is a 3D mode. Many modern ocean models, such as the “MPAS-Ocean” [15], which was developed by Los Alamos National Laboratory and collaborating institutions, use this mode splitting process to design efficient numerical methods for solving the primitive equations. To advance the baroclinic mode, these models use explicit time integration schemes that take advantage of the natural parallelism and ease implementation of explicit stepping. However, there are two methods for integrating the barotropic mode: an explicit-subcycling approach and a semi-implicit solver approach. The explicit-subcycling approach advances the barotropic mode using a smaller time step size to meet the more restrictive CFL condition for the barotropic subsystem [11, 13, 15]. The resulting numerical scheme, known as the split-explicit (SE) method, requires updating the intermediate values during each subcycling step. This approach imposes intensive message exchange among the processors/cores of high-performance computing clusters, which may significantly affect the parallel scalability. Moreover, the SE method is not automati-

cally stable, even when the respective CFL conditions are satisfied, and requires extra effort to ensure the numerical consistency between the two modes due to the mode splitting at the discrete level. On the other hand, the semi-implicit approach evolves the barotropic mode through the use of a linear system solver. Since the semi-implicit schemes are numerically stable, the barotropic mode can be solved using the same time step size as the baroclinic mode uses for explicit stepping, and no substepping is needed. This numerical scheme for the primitive equations is referred to as the implicit-explicit (IMEX) type method. The numerical inconsistency caused by the mode splitting can be resolved much more easily in this case due to the use of the same time step size for both modes. IMEX schemes have been widely studied and used for existing climate models [16–25].

The semi-implicit approach for the barotropic mode requires solving linear systems that gather data across all processors, making direct linear solvers impractical for high-resolution simulations. Therefore, iterative methods for linear systems are often implemented, and significant efforts have been devoted to designing good preconditioners to reduce the number of iterations and improve scalability. The Parallel Ocean Program (POP) [24] uses the Jacobi preconditioned conjugate gradient method as its barotropic solver, which was later improved by introducing a preconditioned Chebyshev-type iterative method combined with an effective block preconditioner based on the error vector propagation method [18]. However, this approach suffers from the estimation of eigenvalue boundaries from the coefficient matrix before the solver iteration [5]. In contrast, the Finite-Volume Sea Ice-Ocean Model (FESOM2) [16] employs the preconditioned pipelined biconjugate gradient stabilized method [26] with a restricted additive Schwarz preconditioner [27], and its performance with unstructured variable-resolution meshes was demonstrated in previous studies. Building upon this work, Kang et al. [5] proposed a semi-implicit approach [28, 29] based on the current MPAS-Ocean framework, replacing the subcycling process with a scalable Crank-Nicolson (CN) solver for the barotropic mode and using a uniform time step for both baroclinic and barotropic modes to evolve the oceanic motion. Extensive numerical tests show that such an approach outperforms the currently implemented SE approach in MPAS-Ocean. However, it is worth noting that this IMEX scheme for the split system is only first-order accurate in time due to the use of inaccurate barotropic forcing in the coupling of the baroclinic and barotropic modes, as shown in a recent study on the MPAS framework [13].

This paper presents an implicit-explicit (IMEX2) time-stepping scheme with second-order accuracy for the baroclinic-barotropic split system of primitive equations. To achieve higher-order accuracy, it is necessary to design the coupling of the baroclinic and barotropic mode solves carefully. In our scheme, we use the strong stability preserving Runge-Kutta (SSPRK) method to achieve high-order temporal accuracy by employing a convex combination of forward-Euler steps. Within the SSPRK framework, we efficiently compute the barotropic forcing term $\bar{G}$ at each stage of the SSPRK stepping [10, 13]. At each time step of our IMEX2 scheme, we first use the second-order SSPRK2 scheme to explicitly advance the whole baroclinic-barotropic split system to compute the baroclinic velocity. Then, we correct the barotropic velocity by using the CN

solver for the barotropic subsystem with a specially interpolated barotropic force based on values from the preceding two forward-Euler baroclinic steppings. As an implicit solver, the CN scheme allows a larger time step compared to explicit solvers, moreover, it is second-order accurate and energy-preserving for the wave equation. Since the equation for the barotropic mode is quite like the shallow water equation, the CN scheme is could be more feasible. Finally, we update the fluid thickness and the sea surface height (SSH) perturbation by coupling the predicted baroclinic and barotropic velocities. For the baroclinic-barotropic split system, there are two ways to calculate the SSH perturbation: one is through the barotropic mode solve, and the other is from the layer thickness. However, these two ways typically produce different results after time and spatial discretization, which can cause inconsistencies and even lead to numerical stability issues [6, 11, 30, 31]. To resolve this issue, two typical approaches are proposed: layer dilation [6] and flux-form reconciliation [11, 31]. Layer dilation adjusts thicknesses of all layers uniformly by implementing a scaling factor on the relevant layer thickness equations, but it reconciles the perturbation while failing to conserve the mass of each layer since the thickness equations are not formulated in flux-divergence form. In contrast, the flux-form reconciliation ensures that the sum of layer fluxes to meet the barotropic flux and the resulting equations remain in flux-divergence form. Thus, the flux-form reconciliation is more commonly used and is the approach utilized in our proposed IMEX2 scheme.

The structure of the paper is organized as follows. In Section 2, we provide an overview of the primitive equations and the associated baroclinic-barotropic mode splitting. Next, in Section 3, we introduce the second-order IMEX scheme for time-stepping of the baroclinic-barotropic split system, which combines the explicit advancing of the baroclinic mode and layer thickness using the SSPRK2 scheme and the semi-implicit solve of the barotropic mode via the CN scheme, along with SSH reconciliation. As followed, we conduct a temporal convergence analysis of the proposed IMEX scheme in Section 4. To demonstrate the performance of the proposed schemes, we carry out the numerical experiments based on two benchmark test cases from the MPAS-Ocean platform in Section 5. Finally, we address some concluding remarks in Section 6.

2 The primitive equations and the baroclinic-barotropic mode splitting

Consider the oceanic motion governed by the primitive equations, which describe the incompressible Boussinesq equations in hydrostatic balance [15, 32, 33]. The model consists of the following equations:

- Thickness equation:

$$\frac{\partial h}{\partial t} + \nabla \cdot (hu) + \frac{\partial}{\partial z}(hw) = 0. \quad (2.1)$$

- Momentum equation:

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{1}{2} \nabla |\mathbf{u}|^2 + (\vec{k} \cdot \nabla \times \mathbf{u}) \mathbf{u}^\perp + f \mathbf{u}^\perp + w \frac{\partial \mathbf{u}}{\partial z} = -\frac{1}{\rho_0} \nabla p + \nu_h \nabla^2 \mathbf{u} + \frac{\partial}{\partial z} \left(\nu_v \frac{\partial \mathbf{u}}{\partial z} \right). \quad (2.2)$$

- Tracer equations:

$$\frac{\partial h\varphi}{\partial t} + \nabla \cdot (h\varphi \mathbf{u}) + \frac{\partial}{\partial z} (h\varphi w) = \nabla \cdot (h\kappa_h \nabla \varphi) + h \frac{\partial}{\partial z} \left(\kappa_v \frac{\partial \varphi}{\partial z} \right). \quad (2.3)$$

- Hydrostatic condition:

$$p = p^s(x, y) + \int_z^{z^s} \rho g dz'. \quad (2.4)$$

- Equation of state:

$$\rho = f_{\text{eos}}(\Theta, S, p). \quad (2.5)$$

Table 1 provides a list of variables and other entities used in these equations and $\mathbf{u}^\perp = \vec{k} \times \mathbf{u}$.

Table 1: List of variables and other entities for the primitive equations (2.1)-(2.5).

Variables	Definition
h	Understood as the thickness of a fluid layer between density ρ and $\rho + \Delta\rho$ where $\Delta\rho$ is any small constant density difference
$\mathbf{u}, w$	Horizontal and vertical velocity
$p, p^s(x, y)$	Pressure and the top surface pressure
Θ	Potential temperature
S	Salinity
ρ, ρ_0	Density and referential density
g	Gravity acceleration
φ	Generic tracer (Θ or S)
z, z^s	Vertical coordinate and the z-location of the top sea surface
ν_h, ν_v	Horizontal and vertical viscosities
κ_h, κ_v	Tracer diffusion coefficients
$\vec{k}$	Unit vector point in the local vertical direction
f	Coriolis parameter
f_{eos}	Equation of state

In order to address the effects of stratification, layered models are often used. These models partition the vertical range into L segments, with each segment representing a layer. The primitive equations are then discretized in the vertical direction, following the approach used in [15, 34]. Specifically, we take the vertical average of the generic variable

ϕ in layer k and denote it as ϕ_k for $k=1,2,\dots,L$. Within each layer, the horizontal velocity $\mathbf{u}_k$, the layer thickness h_k , and the generic tracer variables ϕ_k are situated in the middle of the layer. Meanwhile, the vertical velocity w_k is positioned at the top of each layer in a vertically staggered pattern. Additionally, we adopt simplified notations for some operators in the z -direction [15, Appendix A.3] as follows:

$$\begin{cases} \overline{(\phi)}_k^m = (\phi_k + \phi_{k+1})/2, & \overline{(\phi)}_k^t = (\phi_{k-1} + \phi_k)/2, \\ D_{z^m}(\phi)_k = \frac{\phi_k - \phi_{k+1}}{h_k}, & D_{z^t}(\phi)_k = \frac{\phi_{k-1} - \phi_k}{(\overline{h})_k^t}. \end{cases}$$

Let us discretize the vertical derivative terms as

$$\frac{\partial}{\partial z}(h_k w_k) \approx D_{z^m}(w)_k h_k, \quad w_k \frac{\partial \mathbf{u}_k}{\partial z} \approx \overline{(w D_{z^t}(\mathbf{u}))}_k^m, \quad \frac{\partial}{\partial z}(h_k \phi_k w_k) \approx D_{z^m}(\overline{(\phi)}_k^t w)_k h_k,$$

and the vertical diffusion terms as

$$\frac{\partial}{\partial z}(\nu_v \frac{\partial \mathbf{u}_k}{\partial z}) \approx D_{z^m}(\nu_v D_{z^t}(\mathbf{u}))_k, \quad \frac{\partial}{\partial z}(\kappa_v \frac{\partial \phi_k}{\partial z}) \approx D_{z^m}(\kappa_v D_{z^t}(\phi))_k.$$

Then the multi-layer primitive equations [15] are given as: for $k=1,\dots,L$,

$$\begin{cases} \frac{\partial h_k}{\partial t} + \nabla \cdot (h_k \mathbf{u}_k) + D_{z^m}(w)_k h_k = 0, \end{cases} \quad (2.6)$$

$$\begin{cases} \frac{\partial \mathbf{u}_k}{\partial t} + \frac{1}{2} \nabla |\mathbf{u}_k|^2 + (\vec{k} \cdot \nabla \times \mathbf{u}_k) \mathbf{u}_k^\perp + f \mathbf{u}_k^\perp + \overline{(w D_{z^t}(\mathbf{u}))}_k^m = -\frac{1}{\rho_0} \nabla p_k + \nu_h \nabla^2 \mathbf{u}_k \\ \quad + D_{z^m}(\nu_v D_{z^t}(\mathbf{u}))_k, \end{cases} \quad (2.7)$$

$$\begin{cases} \frac{\partial h_k \phi_k}{\partial t} + \nabla \cdot (h_k \phi_k \mathbf{u}_k) + D_{z^m}(\overline{(\phi)}_k^t w)_k h_k = \nabla \cdot (h_k \kappa_h \nabla \phi_k) + h_k D_{z^m}(\kappa_v D_{z^t}(\phi))_k, \end{cases} \quad (2.8)$$

$$\begin{cases} p_k = p_k^s(x, y) + \sum_{l=1}^{k-1} \rho_l g h_l + \frac{1}{2} \rho_k g h_k, \end{cases} \quad (2.9)$$

$$\begin{cases} \rho_k = f_{\text{eos}}(\Theta_k, S_k, p_k). \end{cases} \quad (2.10)$$

The choice of vertical coordinate has a significant impact on the computation of w_k via (2.6). For instance, in idealized isopycnal vertical coordinates, there is no vertical transport, resulting in w_k being set to zero. In z -level vertical coordinates, all layers have a fixed thickness, except for the top layer ($k=1$), resulting in w_k being zero for all layers except the first. Conversely, in z -star vertical coordinates, the layer thickness is proportional to the SSH, which means that w_k is non-zero for all layers when the ocean is not at rest. Petersen et al. [33] introduced a new quantity called h_k^{ALE} to represent the expected layer thickness, and proposed the arbitrary Lagrangian-Eulerian (ALE) vertical coordinates to obtain w_k in a unified way.

$$w_k = w_{k+1} - \nabla \cdot (h_k \mathbf{u}_k) - \frac{h_k^{\text{ALE}} - h_k}{\Delta t}.$$

We refer the readers to [33] for more details. In our proposed methods and numerical experiments, we consider the z-star vertical coordinates, which is also the default setting in MPAS-Ocean.

Remark 2.1. In Eq. (2.7), the Laplacian form is not the sole formula for horizontal diffusion. Alternative formulations or turbulent closures for horizontal dissipation can be used in place of it. For instance, the hyperviscosity $-\nu_h \nabla^4 \mathbf{u}_k$ is often preferred in global ocean circulation models due to its superior scale-selective dissipation effect.

In this work, our focus is on developing an IMEX scheme to integrate the ocean dynamics part of the primitive equations, which includes the thickness and momentum equations (2.6)-(2.7) coupled with the hydrostatic condition (2.9). These equations are essential components of numerical ocean modeling. However, due to the presence of multiple time scales, it is common practice to separate the fast and slow motions and split the momentum equation into barotropic and baroclinic subsystems. To this end, we define the barotropic and baroclinic velocities as $\bar{\mathbf{u}}$ and $\tilde{\mathbf{u}} = \{\tilde{\mathbf{u}}_k\}_{k=1}^L$, respectively, following previous studies [13, 15, 35]:

$$\begin{cases} \bar{\mathbf{u}} = \sum_{k=1}^L h_k \mathbf{u}_k / \sum_{k=1}^L h_k, \\ \tilde{\mathbf{u}}_k = \mathbf{u}_k - \bar{\mathbf{u}}, \quad k = 1, \dots, L. \end{cases} \quad (2.11)$$

For simplicity of exposition, we also introduce the following quantities:

$$\begin{aligned} T_k^u(\mathbf{u}) &= -\frac{1}{2} \nabla |\mathbf{u}_k|^2 - (\vec{k} \cdot \nabla \times \mathbf{u}_k) \mathbf{u}_k^\perp - \overline{(w D_{z^t}(\mathbf{u}))}_k^m - \frac{1}{\rho_0} \nabla p_k + \nu_h \nabla^2 \mathbf{u}_k, \\ D_k^u(\mathbf{u}) &= D_{z^m}(\nu_v D_{z^t}(\mathbf{u}))_k, \quad T_k^h(h, \mathbf{u}) = -\nabla \cdot (h_k \mathbf{u}_k) - D_{z^m}(w)_k h_k. \end{aligned}$$

Taking the layer-thickness-weighted average of (2.7), one then can rewrite the dynamics of the primitive equations into the following baroclinic-barotropic split system:

$$\text{Baroclinic mode:} \quad \frac{\partial \tilde{\mathbf{u}}_k}{\partial t} = -f \tilde{\mathbf{u}}_k^\perp + T_k^u(\mathbf{u}) + D_k^u(\mathbf{u}) + g \nabla \zeta - \bar{\mathbf{G}}, \quad (2.12a)$$

$$\text{Barotropic mode:} \quad \frac{\partial}{\partial t} \begin{pmatrix} \bar{\mathbf{u}} \\ \zeta \end{pmatrix} = - \begin{pmatrix} f \vec{k} \times \bar{\mathbf{u}} + g \nabla \zeta \\ \nabla \cdot (\bar{\mathbf{u}}(\zeta + H)) \end{pmatrix} + \begin{pmatrix} \bar{\mathbf{G}} \\ 0 \end{pmatrix}, \quad (2.12b)$$

$$\text{Fluid thickness:} \quad \frac{\partial h_k}{\partial t} = T_k^h(h, \mathbf{u}), \quad (2.12c)$$

where ζ denotes the perturbation of sea surface height (SSH), H represents the total column height relative to SSH, and the nonlinear terms in the equations for baroclinic and barotropic modes are contained in the barotropic forcing term $\bar{\mathbf{G}}$. While the baroclinic subsystem (2.12a) and the fluid thickness equations (2.12c) are three-dimensional problems involving all L layers, the barotropic subsystem (2.12b) is a two-dimensional

problem. Explicitly obtaining the formula for $\bar{G}$ is difficult, so an elegant forward-Euler method was proposed in [10] to numerically evaluate it. This method utilizes the layer-thickness-weighted average free condition for the baroclinic mode instead. It should be noted that the SSH perturbation ζ can be derived from the layer thickness by $\zeta = \sum_{k=1}^L h_k - H$, or from the barotropic subsystem (2.12b). These two methods are essentially equivalent in terms of continuous space and time.

3 A second-order IMEX scheme for the baroclinic-barotropic split system

We propose a second-order IMEX (IMEX2) time-stepping scheme for efficiently solving the baroclinic-barotropic split system (2.12) of the primitive equations. Our method is based on the second-order strong stability preserving Rung-Kutta (SSPRK2) method [36, 37] and the linearized Crank-Nicolson (CN) scheme with special treatments for the coupling of the fast and slow modes. Specifically, the baroclinic mode and layer thickness equation are advanced by the SSPRK2 scheme, and the barotropic mode is solved using a linearized CN scheme.

First, let us recap SSPRK2 scheme for a general system of ordinary differential equations (ODEs):

$$\partial_t \mathbf{V} = F(\mathbf{V}), \quad (3.1)$$

where $\mathbf{V}(t)$ is a vector of unknown variables with the initial value $\mathbf{V}(0)$. Given $\mathbf{V}^n$ at the time t_n , the time-step size Δt and $\mathbf{V}^0 = \mathbf{V}(0)$, to find $\mathbf{V}^{n+1}$ at the time $t_{n+1} = t_n + \Delta t$, The SSPRK2 scheme consists of two stages and uses the following convex combinations of first-order accurate forward-Euler steps:

$$\begin{cases} \hat{\mathbf{V}}^{n+1} = \mathbf{V}^n + \Delta t F(\mathbf{V}^n), \\ \mathbf{V}^{n+1} = \frac{1}{2} \mathbf{V}^n + \frac{1}{2} \left(\hat{\mathbf{V}}^{n+1} + \Delta t F(\hat{\mathbf{V}}^{n+1}) \right). \end{cases} \quad (3.2)$$

We note that $\hat{\varphi}$ will denote the predicted intermediate value of the generic variable φ in this paper.

3.1 Forward-Euler stepping for the baroclinic mode

In ocean dynamics, the baroclinic and barotropic modes correspond to different time scales, leading to distinct CFL conditions. Generally, the baroclinic mode permits larger time-step sizes, while the barotropic mode requires smaller time-step sizes or an implicit solver. Therefore, high-order accuracy numerical schemes require careful consideration of the coupling between these modes. The barotropic forcing term, denoted as $\bar{G}$, is a key component of the split system (2.12). In [11], R. Higdon proposed an effective way to compute $\bar{G}$ using the forward-Euler method for solving the baroclinic subsystem

(2.12a). We adopt this approach and summarize it in Eq. (3.3): given $(\mathbf{u}^n, \tilde{\mathbf{u}}^n, \zeta^n, h^n)$, find $(\hat{\mathbf{u}}^{n+1}, \bar{\mathbf{G}}^n)$, such that

$$\begin{cases} \hat{\mathbf{u}}_k^{n+1} = \tilde{\mathbf{u}}_k^n + \Delta t \left(-f \tilde{\mathbf{u}}_k^{n,\perp} + \mathbf{T}_k^u(\mathbf{u}^n) + \mathbf{D}_k^u(\mathbf{u}^n) + g \nabla \zeta^n \right), & k=1, \dots, L, \\ \bar{\mathbf{G}}^n = \frac{1}{\Delta t} \sum_{k=1}^L h_k^n \hat{\mathbf{u}}_k^{n+1} / \sum_{k=1}^L h_k^n, \\ \hat{\mathbf{u}}_k^{n+1} = \hat{\mathbf{u}}_k^{n+1} - \Delta t \bar{\mathbf{G}}^n, & k=1, \dots, L. \end{cases} \quad (3.3)$$

Following [13], we refer Eq. (3.3) as

$$[\hat{\mathbf{u}}^{n+1}, \bar{\mathbf{G}}^n] = \text{Baroclinic_FEuler}(\mathbf{u}^n, \tilde{\mathbf{u}}^n, \zeta^n, h^n, \Delta t). \quad (3.4)$$

In some ocean modeling scenarios, the vertical mesh sizes are usually much smaller than the horizontal ones, leading to a stiffer equation of vertical motion compared to that of horizontal motion. To address this issue, a vertical mixing technique can be applied to treat the vertical diffusion term $\mathbf{D}_k^u(\mathbf{u})$ implicitly in (2.12a). This technique essentially applies a first-order operator splitting approach. Considering boundary conditions like bottom drag, it is preferable to perform vertical mixing on the entire velocity field $\mathbf{u}$ instead of solely on the baroclinic velocity $\tilde{\mathbf{u}}$. We summarize our approach in Eq. (3.5): given $(\mathbf{u}^n, \tilde{\mathbf{u}}^n, \zeta^n, h^n)$, find $(\hat{\mathbf{u}}^{n+1}, \bar{\mathbf{G}}^n)$, such that:

$$\begin{cases} \hat{\mathbf{u}}_k^{n+1} = \tilde{\mathbf{u}}_k^n + \Delta t \left(-f \tilde{\mathbf{u}}_k^{n,\perp} + \mathbf{T}_k^u(\mathbf{u}^n) + g \nabla \zeta^n \right), & k=1, \dots, L, \\ \bar{\mathbf{G}}^n = \frac{1}{\Delta t} \sum_{k=1}^L h_k^n \hat{\mathbf{u}}_k^{n+1} / \sum_{k=1}^L h_k^n, \\ \hat{\mathbf{u}}_k^{n+1} = \hat{\mathbf{u}}_k^{n+1} - \Delta t \bar{\mathbf{G}}^n, & k=1, \dots, L, \\ \hat{\mathbf{u}}^{n+1} - \Delta t \mathbf{D}^u(\hat{\mathbf{u}}^{n+1}) = \hat{\mathbf{u}}_k^{n+1} + \bar{\mathbf{u}}^n, \\ \hat{\mathbf{u}}_k^{n+1} = \hat{\mathbf{u}}^{n+1} - \bar{\mathbf{u}}^n. \end{cases} \quad (3.5)$$

Following [13], we refer Eq. (3.5) as

$$[\hat{\mathbf{u}}^{n+1}, \bar{\mathbf{G}}^n] = \text{Baroclinic_FEuler_Mixing}(\mathbf{u}^n, \tilde{\mathbf{u}}^n, \zeta^n, h^n, \Delta t). \quad (3.6)$$

To advance the baroclinic subsystem (2.12a), we use the forward-Euler scheme with vertical mixing, thereby solving the vertical diffusion equation involving $\hat{\mathbf{u}}^{n+1}$ (i.e., Eq. (3.5) instead of Eq. (3.3)). After spatial discretization, the resulting linear system comprises blocks of tridiagonal coefficient submatrices, which can be solved efficiently using a tridiagonal matrix algorithm [38] along each independent vertical direction. The Community

Vertical Mixing Project (CVMix) [39] shows that the vertical mixing has no significant effect on explicit time-stepping performance. However, according to the study presented in [13], when the impact of the vertical diffusion term $D_k^u(\mathbf{u})$ on the entire system is substantial, the utilization of this technique can significantly decrease the high-order accuracy of time-stepping schemes, reducing it to as low as 1 in the worst-case scenario. From now on, we will repeatedly use Eq. (3.3) to develop our IMEX2 time-stepping scheme and compute the required values of the barotropic forcing term $\bar{G}$. In some cases where needed, we may use Eq. (3.5) instead.

3.2 Linearized Crank-Nicolson scheme for the barotropic mode

Let us first apply the CN scheme to the barotropic subsystem (2.12b). That is, given $(\bar{\mathbf{u}}^n, \zeta^n)$, find $(\bar{\mathbf{u}}^{n+1}, \zeta^{n+1})$ such that

$$\begin{pmatrix} \bar{\mathbf{u}}^{n+1} \\ \zeta^{n+1} \end{pmatrix} = \begin{pmatrix} \bar{\mathbf{u}}^n \\ \zeta^n \end{pmatrix} - \Delta t \begin{pmatrix} f\vec{k} \times \bar{\mathbf{u}}^{n+\frac{1}{2}} + g\nabla\zeta^{n+\frac{1}{2}} \\ \nabla \cdot (\bar{\mathbf{u}}^{n+\frac{1}{2}}(\zeta^{n+\frac{1}{2}} + H)) \end{pmatrix} + \Delta t \begin{pmatrix} \bar{G}^{n+\frac{1}{2}} \\ 0 \end{pmatrix},$$

where

$$\bar{\mathbf{u}}^{n+\frac{1}{2}} = \frac{\bar{\mathbf{u}}^{n+1} + \bar{\mathbf{u}}^n}{2}, \quad \zeta^{n+\frac{1}{2}} = \frac{\zeta^{n+1} + \zeta^n}{2}.$$

Note that the value of $\bar{G}^{n+\frac{1}{2}}$ plays an important role in coupling the baroclinic and barotropic modes, and needs to be carefully chosen for the discrete split system. Since the equation about ζ^{n+1} is nonlinear, we linearize the nonlinear term by

$$\nabla \cdot (\bar{\mathbf{u}}^{n+\frac{1}{2}}(\zeta^{n+\frac{1}{2}} + H)) \approx \frac{1}{2} \nabla \cdot (\bar{\mathbf{u}}^n(\zeta^{n+1} + H)) + \frac{1}{2} \nabla \cdot (\bar{\mathbf{u}}^{n+1}(\zeta^n + H)), \quad (3.7)$$

then the linearized barotropic mode solve can be represented as follows:

$$\begin{pmatrix} \bar{\mathbf{u}}^{n+1} \\ \zeta^{n+1} \end{pmatrix} = \begin{pmatrix} \bar{\mathbf{u}}^n \\ \zeta^n \end{pmatrix} - \Delta t \begin{pmatrix} f\vec{k} \times \bar{\mathbf{u}}^{n+\frac{1}{2}} + g\nabla\zeta^{n+\frac{1}{2}} \\ \frac{1}{2} \nabla \cdot (\bar{\mathbf{u}}^n(\zeta^{n+1} + H)) + \frac{1}{2} \nabla \cdot (\bar{\mathbf{u}}^{n+1}(\zeta^n + H)) \end{pmatrix} + \Delta t \begin{pmatrix} \bar{G}^{n+\frac{1}{2}} \\ 0 \end{pmatrix}. \quad (3.8)$$

Let us refer the scheme (3.8) as

$$[\bar{\mathbf{u}}^{n+1}] = \text{Barotropic.CN}(\bar{\mathbf{u}}^n, \zeta^n, \bar{G}^1, \bar{G}^2, \Delta t) \quad (3.9)$$

with $\bar{G}^{n+\frac{1}{2}} = \frac{\bar{G}^1 + \bar{G}^2}{2}$, where the values of $\bar{G}^1$ and $\bar{G}^2$ will be explicitly specified in the proposed IMEX2 scheme.

3.3 The IMEX2 scheme coupling the baroclinic and barotropic mode solves

The proposed IMEX2 scheme for solving the split system (2.12) consisting of the baroclinic and barotropic modes requires three steps at each time-stepping from t_n to t_{n+1} . Step 2 uses the CN scheme to solve the barotropic subsystem (2.12b) in order to obtain the correct barotropic velocity. To compute the values of $\bar{G}_0^n$ and $\bar{G}_1^n$ used in the CN scheme, two forward-Euler baroclinic solves are performed in Step 1. At each time-stepping from t_n to t_{n+1} , the proposed IMEX2 scheme is summarized in Algorithm 1 on the next page.

3.4 Sea surface height reconciliation in flux-form

As aforementioned, the SSH perturbation ζ resulting from mode splitting can be obtained through two methods: from the barotropic CN solver or from the layer thickness. However, this can lead to model inconsistency errors due to the splitting and the implicit inclusion of the fast mode in the layer thickness. Therefore, to address this issue and further stabilize the fast mode, we will implement a flux-form SSH reconciliation process [11, 13] within the proposed IMEX2 schemes. This process involves matching the value of ζ computed from the fluid layer thickness with that produced from the barotropic CN scheme at each stage.

Firstly, it is worth noting that the SSH perturbations obtained from the two barotropic CN solutions ((3.10b) and (3.12a), respectively) can be represented as follows: for $i = 1, 2$,

$$\zeta_i = \zeta^n - \Delta t \nabla \cdot \mathbf{F}_i^\zeta$$

with the accumulated flux

$$\mathbf{F}_i^\zeta = \begin{cases} \frac{1}{2} \bar{\mathbf{u}}^n (\zeta^{n+1} + H) + \frac{1}{2} \hat{\mathbf{u}}^{n+1} (\zeta^n + H), & i = 1, \\ \frac{1}{2} \bar{\mathbf{u}}^n (\zeta^{n+1} + H) + \frac{1}{2} \bar{\mathbf{u}}^{n+1} (\zeta^n + H), & i = 2. \end{cases}$$

On the other hand, the SSH perturbations resulting from the baroclinic velocity obtained through SSPRK2 are expressed as the summation of layer-thickness perturbations in (3.10e) and (3.13b), that is: for $i = 1, 2$,

$$\zeta_i = \zeta^n - \Delta t \nabla \cdot \mathbf{F}_i^h,$$

with

$$\mathbf{F}_i^h = \begin{cases} \sum_{k=1}^L \mathbf{u}_k^n h_k^n, & i = 1, \\ \sum_{k=1}^L \frac{\mathbf{u}_k^n h_k^n + \mathbf{u}_k^{n+1} \hat{h}_k^{n+1}}{2}, & i = 2. \end{cases}$$

To compensate for the flux deficits $D_i^e = \mathbf{F}_i^\zeta - \mathbf{F}_i^h$ between each pair of stages, $i = 1, 2$, we modify the calculations of the layer thickness at each stage. Specifically, we adjust the layer thickness flux by explicitly assigning the deficit D_i^e in a layer thickness-weighted

Algorithm 1: IMEX2 for baroclinic-barotropic system

Input: $\mathbf{u}^n, \zeta^n, h^n, \Delta t$

Output: $\mathbf{u}^{n+1}, h^{n+1}, \zeta^{n+1}$

Preprocessing: Compute $\bar{\mathbf{u}}^n$ and $\hat{\mathbf{u}}^n$ using the splitting formula (2.11).

Step 1. To compute the baroclinic velocity $\hat{\mathbf{u}}^{n+1}$ for the baroclinic-barotropic system (2.12) at time t_{n+1} , we use the SSPRK2 method with a time step Δt starting from $\mathbf{u}^n$. Additionally, we advance the fluid thickness for the first stage in the same manner. To assist in this process, we predict one intermediate barotropic velocity using the linearized CN scheme with Δt :

/* Stage 1 of SSPRK2 for evolving the whole system */

$$\left[\hat{\mathbf{u}}^{n+1}, \bar{\mathbf{G}}_0^n \right] = \text{Baroclinic_FEuler}(\mathbf{u}^n, \hat{\mathbf{u}}^n, \zeta^n, h^n, \Delta t). \quad (3.10a)$$

$$\left[\hat{\mathbf{u}}^{n+1} \right] = \text{Barotropic_CN}(\bar{\mathbf{u}}^n, \zeta^n, \bar{\mathbf{G}}_0^n, \bar{\mathbf{G}}_0^n, \Delta t). \quad (3.10b)$$

$$\hat{\mathbf{u}}_k^{n+1} = \hat{\mathbf{u}}^{n+1} + \hat{\mathbf{u}}_k^{n+1}. \quad (3.10c)$$

$$\begin{cases} \hat{h}_k^{n+1} = h_k^n + \Delta t \mathbf{T}_k^h(h^n, \mathbf{u}^n), \\ \hat{\zeta}^{n+1} = \sum_{k=1}^L \hat{h}_k^{n+1} - H. \end{cases} \quad (3.10d)$$

$$\begin{cases} \hat{h}_k^{n+1} = h_k^n + \Delta t \mathbf{T}_k^h(h^n, \mathbf{u}^n), \\ \hat{\zeta}^{n+1} = \sum_{k=1}^L \hat{h}_k^{n+1} - H. \end{cases} \quad (3.10e)$$

/* Stage 2 of SSPRK2 for baroclinic velocity */

$$\begin{cases} \left[\hat{\mathbf{u}}^{n+2}, \bar{\mathbf{G}}_1^n \right] = \text{Baroclinic_FEuler}(\hat{\mathbf{u}}^{n+1}, \hat{\mathbf{u}}^{n+1}, \hat{\zeta}^{n+1}, \hat{h}^{n+1}, \Delta t), \\ \hat{\mathbf{u}}^{n+1} = \frac{1}{2}(\hat{\mathbf{u}}^n + \hat{\mathbf{u}}^{n+2}). \end{cases} \quad (3.11a)$$

$$\begin{cases} \hat{\mathbf{u}}^{n+1} = \frac{1}{2}(\hat{\mathbf{u}}^n + \hat{\mathbf{u}}^{n+2}). \end{cases} \quad (3.11b)$$

Step 2. Re-advance the barotropic subsystem (2.12b) using the barotropic forcings $\bar{\mathbf{G}}_0^n$ and $\bar{\mathbf{G}}_1^n$ from t_n via the linearized CN scheme to compute the correct barotropic velocity $\bar{\mathbf{u}}^{n+1}$:

$$\left[\bar{\mathbf{u}}^{n+1} \right] = \text{Barotropic_CN}(\bar{\mathbf{u}}^n, \zeta^n, \bar{\mathbf{G}}_0^n, \bar{\mathbf{G}}_1^n, \Delta t). \quad (3.12a)$$

$$\mathbf{u}_k^{n+1} = \bar{\mathbf{u}}^{n+1} + \hat{\mathbf{u}}_k^{n+1}. \quad (3.12b)$$

Step 3. Continue to advance the fluid thickness to achieve h_k^{n+1} and update the SSH perturbation ζ^{n+1} :

/* Stage 2 of SSPRK2 for fluid thickness */

$$\begin{cases} \hat{h}_k^{n+2} = \hat{h}_k^{n+1} + \Delta t \mathbf{T}_k^h(\hat{h}^{n+1}, \mathbf{u}^{n+1}), \quad h_k^{n+1} = \frac{1}{2}(h_k^n + \hat{h}_k^{n+2}), \\ \zeta^{n+1} = \sum_{k=1}^L h_k^{n+1} - H. \end{cases} \quad (3.13a)$$

$$\begin{cases} \zeta^{n+1} = \sum_{k=1}^L h_k^{n+1} - H. \end{cases} \quad (3.13b)$$

manner to ensure computational efficiency. This leads to defining transport velocity adjustments as:

$$\mathbf{u}_i^{A,n+1} = \begin{cases} \frac{D_1^e}{\sum_{k=1}^L h_k^n}, & i=1, \\ \frac{2D_2^e}{\sum_{k=1}^L h_k^{n+1}}, & i=2, \end{cases} \quad (3.14)$$

where the appearing factor of 2 when $i=2$ is a result of the final layer thickness averaging step. With this, our SSH reconciliation process is completed by modifying the layer thickness updates (3.10d) and (3.13a) to

$$\hat{h}_k^{n+1} = h_k^n + \Delta t \mathbf{T}_k^h(h^n, \mathbf{u}^n + \mathbf{u}_1^{A,n+1}) \quad (3.15)$$

and

$$\hat{h}_k^{n+2} = \hat{h}_k^{n+1} + \Delta t \mathbf{T}_k^h(\hat{h}^{n+1}, \mathbf{u}^{n+1} + \mathbf{u}_2^{A,n+1}), \quad h_k^{n+1} = \frac{1}{2}(h_k^n + \hat{h}_k^{n+2}) \quad (3.16)$$

respectively.

Remark 3.1. The above reconciliation follows almost the same idea of [13]. The difference mainly lies in the calculation of $\mathbf{F}_i^\zeta$, which is much simpler than the original one in [13] – it only needs to accumulate the fluxes from two time levels while it has to sum up all the fluxes from the whole sub-cycling levels in [13].

4 Temporal error analysis

In this section, we analyze and discuss the temporal errors of the proposed IMEX2 scheme for solving the semi-discrete-in-time case of the baroclinic-barotropic split system (2.12) without considering spatial discretization. We also assume that the solutions are sufficiently smooth, and neither vertical mixing nor SSH reconciliation is applied. Based on these assumptions, we can derive the following error estimate for our IMEX2 scheme.

Theorem 4.1. *The IMEX2 scheme given by Algorithm 1 is of second-order accuracy in time for solving the semi-discrete-in-time case of the baroclinic-barotropic split system (2.12), under the assumptions that the solutions are sufficiently smooth and that neither vertical mixing nor SSH reconciliation is applied.*

Proof. First, let us estimate the linearization part (3.7) in the CN barotropic mode solve.

If we apply the Taylor's expansion to the whole terms at $t_{n+1/2}$, it has

$$\begin{aligned}
& \frac{1}{2} \nabla \cdot (\bar{\mathbf{u}}^n (\zeta^{n+1} + H)) + \frac{1}{2} \nabla \cdot (\bar{\mathbf{u}}^{n+1} (\zeta^n + H)) \\
&= \frac{1}{2} \nabla \cdot \left((\bar{\mathbf{u}}^{n+1/2} - \frac{\Delta t}{2} \bar{\mathbf{u}}_t^{n+1/2} + \mathcal{O}(\Delta t^2)) (\zeta^{n+1/2} + \frac{\Delta t}{2} \zeta_t^{n+1/2} + \mathcal{O}(\Delta t^2) + H) \right) \\
& \quad + \frac{1}{2} \nabla \cdot \left((\bar{\mathbf{u}}^{n+1/2} + \frac{\Delta t}{2} \bar{\mathbf{u}}_t^{n+1/2} + \mathcal{O}(\Delta t^2)) (\zeta^{n+1/2} - \frac{\Delta t}{2} \zeta_t^{n+1/2} + \mathcal{O}(\Delta t^2) + H) \right) \\
&= \nabla \cdot (\bar{\mathbf{u}}^{n+1/2} (\zeta^{n+1/2} + H)) + \frac{\Delta t}{4} \nabla \cdot (\bar{\mathbf{u}}^{n+1/2} \zeta_t^{n+1/2}) - \frac{\Delta t}{4} \nabla \cdot (\bar{\mathbf{u}}_t^{n+1/2} (\zeta^{n+1/2} + H)) \\
& \quad - \frac{\Delta t}{4} \nabla \cdot (\bar{\mathbf{u}}^{n+1/2} \zeta_t^{n+1/2}) + \frac{\Delta t}{4} \nabla \cdot (\bar{\mathbf{u}}_t^{n+1/2} (\zeta^{n+1/2} + H)) + \mathcal{O}(\Delta t^2) \\
&= \nabla \cdot (\bar{\mathbf{u}}^{n+1/2} (\zeta^{n+1/2} + H)) + \mathcal{O}(\Delta t^2).
\end{aligned}$$

That is,

$$\nabla \cdot (\bar{\mathbf{u}}^{n+1/2} (\zeta^{n+1/2} + H)) = \frac{1}{2} \nabla \cdot (\bar{\mathbf{u}}^n (\zeta^{n+1} + H)) + \frac{1}{2} \nabla \cdot (\bar{\mathbf{u}}^{n+1} (\zeta^n + H)) + \mathcal{O}(\Delta t^2). \quad (4.1)$$

Next, let us turn to the baroclinic mode solve. It is easy to show that if we add $\mathcal{O}(\Delta t^3)$ terms to the forward-Euler steps in the classic SSPRK2 scheme, that is,

$$\begin{cases} \hat{\mathbf{V}}^{n+1} = \mathbf{V}^n + \Delta t F(\mathbf{V}^n) + \mathcal{O}(\Delta t^3), \\ \mathbf{V}^{n+1} = \frac{1}{2} \mathbf{V}^n + \frac{1}{2} (\hat{\mathbf{V}}^{n+1} + \Delta t F(\hat{\mathbf{V}}^{n+1}) + \mathcal{O}(\Delta t^3)), \end{cases}$$

then the resulting scheme is still second-order accurate in time, which is a key observation for our proof. In the sequel, we mark the values obtained by following the classic SSPRK2 scheme with the subscript "RK2".

At the first stage, the barotropic mode solve (3.10b), which is the modified CN scheme with the value of $\bar{\mathbf{G}}$ at $t=t_n$. Hence, we only have the first-order accurate approximations, that is,

$$\hat{\bar{\mathbf{u}}}^{n+1} = \bar{\mathbf{u}}^n - \Delta t F^n + \mathcal{O}(\Delta t^2),$$

which is equivalent to the forward-Euler with the extra term $\mathcal{O}(\Delta t^2)$. Then it holds for the predicted whole velocity $\hat{\mathbf{u}}^{n+1}$ that

$$\hat{\mathbf{u}}^{n+1} = \hat{\bar{\mathbf{u}}}^{n+1} + \hat{\mathbf{u}}^{n+1} = \hat{\bar{\mathbf{u}}}^{n+1}_{\text{RK2}} + (\hat{\bar{\mathbf{u}}}^{n+1}_{\text{RK2}} + \mathcal{O}(\Delta t^2)) = \hat{\mathbf{u}}^{n+1}_{\text{RK2}} + \mathcal{O}(\Delta t^2).$$

At the second stage, (3.11a) is performed and we obtain

$$\begin{aligned}
\hat{\bar{\mathbf{u}}}^{n+2} &= \text{Baroclinic_FEuler}(\hat{\mathbf{u}}^{n+1}_{\text{RK2}} + \mathcal{O}(\Delta t^2), \hat{\mathbf{u}}^{n+1}_{\text{RK2}}, \hat{\zeta}^{n+1}_{\text{RK2}}, \hat{h}^{n+1}_{\text{RK2}}, \Delta t) = \hat{\bar{\mathbf{u}}}^{n+2}_{\text{RK2}} + \mathcal{O}(\Delta t^3), \\
\hat{\mathbf{u}}^{n+1} &= \frac{1}{2} (\hat{\mathbf{u}}^n + \hat{\bar{\mathbf{u}}}^{n+2}),
\end{aligned}$$

where $\mathcal{O}(\Delta t^3)$ comes from the fact that $\hat{\mathbf{u}}_{\text{RK2}}^{n+1} + \mathcal{O}(\Delta t^2)$ is a part of the flux term that is then multiplied by Δt again. Therefore, the baroclinic velocity $\hat{\mathbf{u}}^{n+1}$ obtained by the proposed IMEX2 is of second-order accuracy in time. Next, let us move back to the barotropic mode. The proposed IMEX2 recomputes the fast mode with the values at t_n and the predicted terms $\bar{\mathbf{G}}_0^n$ and $\bar{\mathbf{G}}_1^n$. If we substitute the real solutions into the CN scheme, it will have the $\mathcal{O}(\Delta t^3)$ truncation term, i.e.,

$$\begin{pmatrix} \bar{\mathbf{u}}(t_{n+1}) \\ \zeta(t_{n+1}) \end{pmatrix} = \begin{pmatrix} \bar{\mathbf{u}}(t_n) \\ \zeta(t_n) \end{pmatrix} - \Delta t \begin{pmatrix} f\vec{\mathbf{k}} \times \frac{\bar{\mathbf{u}}(t_{n+1}) + \bar{\mathbf{u}}(t_n)}{2} + g\nabla \frac{\zeta(t_{n+1}) + \zeta(t_n)}{2} \\ \nabla \cdot \left(\frac{\bar{\mathbf{u}}(t_{n+1}) + \bar{\mathbf{u}}(t_n)}{2} \left(\frac{\zeta(t_{n+1}) + \zeta(t_n)}{2} + H \right) \right) \end{pmatrix} \\ + \Delta t \begin{pmatrix} \frac{\bar{\mathbf{G}}(t_{n+1}) + \bar{\mathbf{G}}(t_n)}{2} \\ 0 \end{pmatrix} + \mathcal{O}(\Delta t^3),$$

where $\bar{\mathbf{u}}(t)$ and $\zeta(t)$ denote the real solutions at time t . Furthermore, by combining (4.1), we get

$$\begin{pmatrix} \bar{\mathbf{u}}(t_{n+1}) \\ \zeta(t_{n+1}) \end{pmatrix} = \begin{pmatrix} \bar{\mathbf{u}}(t_n) \\ \zeta(t_n) \end{pmatrix} - \Delta t \begin{pmatrix} f\vec{\mathbf{k}} \times \frac{\bar{\mathbf{u}}(t_{n+1}) + \bar{\mathbf{u}}(t_n)}{2} + g\nabla \frac{\zeta(t_{n+1}) + \zeta(t_n)}{2} \\ \frac{1}{2} \nabla \cdot (\bar{\mathbf{u}}(t_n)(\zeta(t_{n+1}) + H)) + \frac{1}{2} \nabla \cdot (\bar{\mathbf{u}}(t_{n+1})(\zeta(t_n) + H)) \end{pmatrix} \\ + \Delta t \begin{pmatrix} \frac{\bar{\mathbf{G}}(t_{n+1}) + \bar{\mathbf{G}}(t_n)}{2} \\ 0 \end{pmatrix} + \mathcal{O}(\Delta t^3).$$

Comparing to the Barotropic-CN scheme (3.12a), one see that $\frac{\bar{\mathbf{G}}(t_{n+1}) + \bar{\mathbf{G}}(t_n)}{2}$ is approximated by $\frac{\bar{\mathbf{G}}_1^n + \bar{\mathbf{G}}_0^n}{2}$. Since the approximations $\bar{\mathbf{G}}_1^n$ and $\bar{\mathbf{G}}_0^n$ are first-order accurate approximations to $\bar{\mathbf{G}}(t_{n+1})$ and $\bar{\mathbf{G}}(t_n)$, thus

$$\frac{\bar{\mathbf{G}}(t_{n+1}) + \bar{\mathbf{G}}(t_n)}{2} = \frac{\bar{\mathbf{G}}_1^n + \bar{\mathbf{G}}_0^n}{2} + \mathcal{O}(\Delta t^2).$$

The second substep of the SSPRK2 scheme given by Eq. (3.12a) has a third-order truncation error, thereby providing a second-order accuracy in time. We can rewrite the update for the layer thickness h compactly as follows:

$$h^{n+1} = h^n - \frac{\Delta t}{2} \left(\mathbf{T}_k^h(h^n, \mathbf{u}^n) + \mathbf{T}_k^h(\hat{h}^{n+1}, \mathbf{u}^{n+1}) \right). \quad (4.2)$$

Here, we use the second-order accurate solution $\mathbf{u}^{n+1}$ instead of the first-order accurate velocity $\hat{\mathbf{u}}_{\text{RK2}}^{n+1}$, resulting in a difference of $\mathcal{O}(\Delta t^2)$ between $\mathbf{u}^{n+1}$ and $\hat{\mathbf{u}}_{\text{RK2}}^{n+1}$. Additionally, following the classic result of SSPRK2, we obtain a third-order truncation error in time for $\hat{h}_{\text{RK2}}^{n+1}$ given by

$$\hat{h}_{\text{RK2}}^{n+1} = h^n - \frac{\Delta t}{2} \left(\mathbf{T}_k^h(h^n, \mathbf{u}^n) + \mathbf{T}_k^h(\hat{h}_{\text{RK2}}^{n+1}, \hat{\mathbf{u}}_{\text{RK2}}^{n+1}) \right), \quad (4.3)$$

where $|\tilde{h}_{\text{RK2}}^{n+1} - h(t_{n+1})| = \mathcal{O}(\Delta t^3)$. Furthermore,

$$|h^{n+1} - \tilde{h}_{\text{RK2}}^{n+1}| = \left| \frac{\Delta t}{2} \left(\mathbf{T}_k^h(\hat{h}^{n+1}, \mathbf{u}^{n+1}) - \mathbf{T}_k^h(\hat{h}_{\text{RK2}}^{n+1}, \hat{\mathbf{u}}_{\text{RK2}}^{n+1}) \right) \right| = \mathcal{O}(\Delta t^3). \quad (4.4)$$

Noticing $\hat{h}^{n+1} = \hat{h}_{\text{RK2}}^{n+1}$ and combining the above three equations (4.2), (4.3) and (4.4), it holds that h^{n+1} is also of second-order accuracy in time. $\square$

5 Numerical experiments

We implemented the proposed IMEX2 time-stepping scheme for the primitive equations, utilizing the Trilinos [40] and MPAS-Ocean [41] framework. The spatial discretization of the baroclinic-barotropic split system (2.12) is achieved through the TRiSK scheme, which is a finite volume approximation specially designed for unstructured, locally orthogonal dual meshes [42–44]. This ensures the preservation of physical properties, such as mass conservation, total energy conservation, and potential vorticity conservation of the space-continuous system in its space-discrete counterpart. For the linearized CN scheme for the barotropic mode, we use the Trilinos AztecOO package [45] to solve the resulting linear systems at each time step, specifically, we choose the preconditioned GMRES [46] (with the built-in ILU preconditioner "AZ_ilu" in its default setting [45]: the level of graph fill-in is set as 0 and drop tolerance is 0.0) as the iterative linear solver. The maximum number of GMRES iterations is set to be 100 and the tolerance for the residual $1.0\text{E-}10$ except stated otherwise. All experiments are performed on the BSCC-A6 super computing system from China Science and Technology Cloud (CSTCloud), which is equipped with AMD 7452@2.35GHz processors (each consists of 256 G memory and 64 cores).

To evaluate the accuracy and performance of the proposed IMEX2 scheme, we will conduct two benchmark tests from the MPAS-Ocean platform (Version 7.0): the baroclinic eddies test and the global ocean test. In both cases, the bottom drag is considered as a bottom boundary condition, which is represented by the following equation:

$$\lim_{z \rightarrow z_{\text{bot}}} \nu_v \frac{\partial \mathbf{u}}{\partial z} = c_{\text{drag}} |\mathbf{u}| \mathbf{u}, \quad (5.1)$$

where $|\mathbf{u}|$ represents the velocity magnitude, c_{drag} is the bottom drag coefficient, and z_{bot} is the z -location of the ocean bottom. To assess the accuracy, we will measure the relative l_2 error for the surface velocity $\mathbf{u}_1$ and the fluid thickness of the top layer h_1 at the terminal time T using the following formula:

$$\frac{\|\mathbf{u}_1^t - \mathbf{u}_1^r\|_2}{\|\mathbf{u}_1^r\|_2} \quad \text{and} \quad \frac{\|h_1^t - h_1^r\|_2}{\|h_1^r\|_2},$$

where $\mathbf{u}_1^r$ and h_1^r are the reference values, and $\mathbf{u}_1^t$ and h_1^t are the testing values. The l_2 -norm of the vector is denoted by $\|\cdot\|_2$.

5.1 The baroclinic eddies test case with twenty layers

We begin by considering the idealized baroclinic eddies test case [15,41] for the primitive equations to test our proposed method. The experiment involves a spin-down scenario without any external forcing, including surface wind or buoyancy forcing. The simulation domain is a horizontally periodic channel with a latitudinal extent of 440 km and a longitudinal extent of 160 km. The domain has a flat bottom with a vertical depth of 1 km and is located on an f-plane [4]. The Coriolis parameter for the experiment is $f = 1.2 \times 10^{-4} \text{ s}^{-1}$, where "s" denotes second. The initial (potential) temperature decreases downward in the meridional direction. Additionally, a cosine-shaped temperature perturbation with a wavelength of 120 km in the zonal direction is applied to induce the baroclinic instability. To partition the horizontal domain, we employ a 10-km-resolution SCVT mesh [47] and divide the vertical domain into 20 layers. Each layer comprises 3,920 cells, 11,840 edges, and 7,920 vertices. We set the horizontal viscosity to $\nu_h = 10$ and the vertical viscosity to $\nu_v = 1.0\text{E-}4$. Since the vertical diffusion has a negligible impact, we utilize Eq. (3.3) ("Baroclinic.FEuler") to advance the baroclinic mode in the IMEX2 scheme.

We set the terminal time $T = 4096 \text{ s}$ and numerically simulate the velocity and layer thickness using the proposed IMEX2 scheme with different time step sizes ($\Delta t = 32, 16, 8, 4$, and 2 s). The reference solution is generated using the IMEX2 scheme with a much smaller time step size $\Delta t = 0.25 \text{ s}$. The results of relative l_2 errors and convergence rates at the terminal time are reported in Table 2, where we can see the almost exact second-order accuracy for this test case. To further test the stability of our IMEX2 scheme, we conduct a 20-day simulation with $\Delta t = 256 \text{ s}$ ($\Delta_{\text{btr}} t \leq 60 \text{ s}$ is allowed for the barotropic mode in SE schemes [13,41]), in which two different bottom drag coefficients $c_{\text{drag}} = 1.0\text{E-}2$ and $c_{\text{drag}} = 1.0\text{E-}3$ are tested. For these simulations, we also combine the MPAS-Ocean temperature marching module into our IMEX2 scheme. Snapshots of the initial surface temperature and values on Day 5, 10, 15 and 20 are shown in Fig. 1, where the top row is associated with the case of $c_{\text{drag}} = 1.0\text{E-}2$ and the bottom row with $c_{\text{drag}} = 1.0\text{E-}3$. As expected, the eddies generated by $c_{\text{drag}} = 1.0\text{E-}3$ escape further away from the interface since it provides the weaker bottom drag.

Table 2: Numerical results on relative l_2 errors and convergence rates of u_1 and h_1 at the terminal time $T = 4096 \text{ s}$ for the baroclinic eddies test case produced by IMEX2 scheme.

Δt	Velocity		Layer thickness	
	Error	Rate	Error	Rate
32	7.886E-03	-	3.301E-06	-
16	2.723E-03	1.53	1.222E-06	1.43
08	7.133E-04	1.93	3.132E-07	1.96
04	1.793E-04	1.99	7.831E-08	2.00
02	4.457E-05	2.01	1.947E-08	2.01

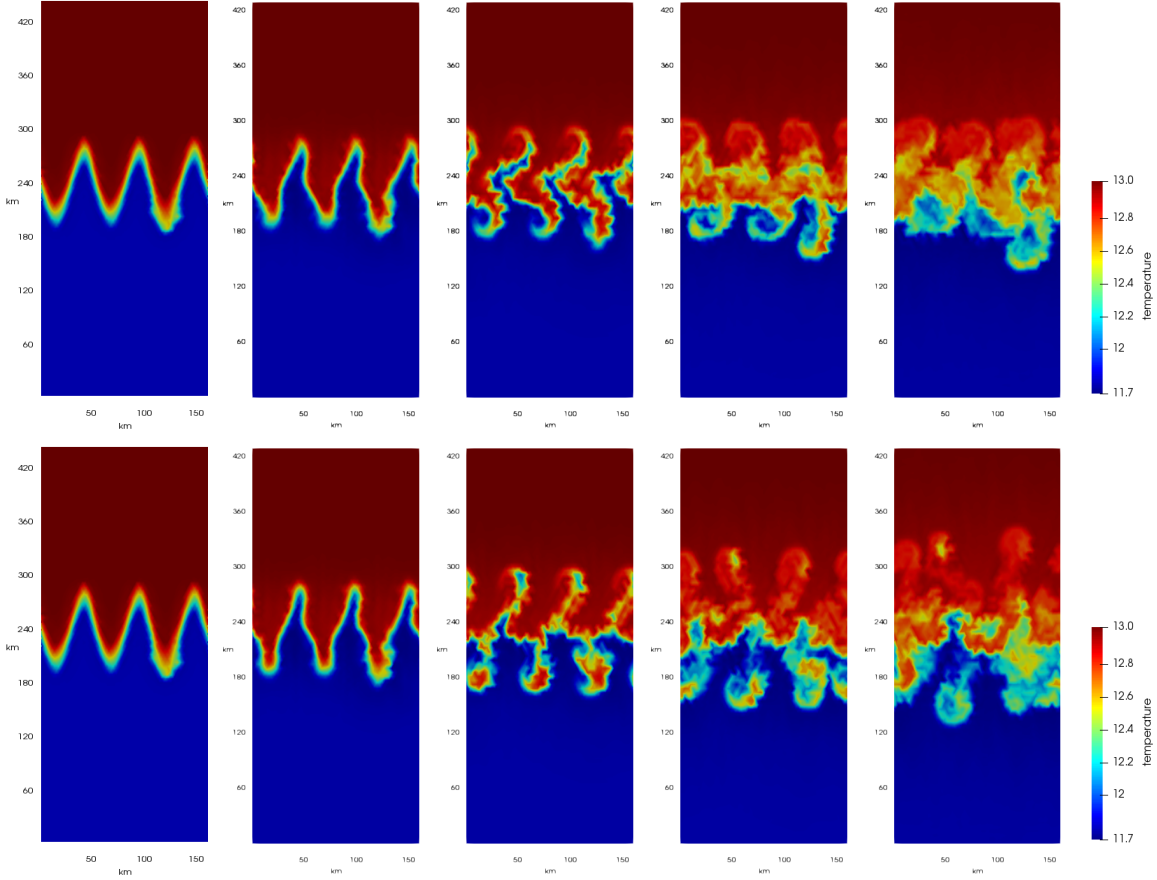


Figure 1: The simulated surface temperatures for the baroclinic eddies test case using the IMEX2 scheme (without vertical mixing) with a time step of $\Delta t = 256$ s are shown from left to right in the results at the initial time and Day 5, 10, 15, and 20. Top: $c_{\text{drag}} = 0.01$; bottom: $c_{\text{drag}} = 0.001$.

5.2 The global ocean test case with one hundred layers

In this example, we consider the global ocean configuration from the MPAS-Ocean platform. The horizontal mesh is a quasi-uniform SCVT mesh, denoted as "QU240", with a cell width of 240 km. The mesh consists of 100 layers, each containing 7,234 cells, 22,736 edges, and 15,459 vertices. In this test case, the hyperviscosity is implemented using the horizontal biharmonic operator with the coefficient $\nu_h = 2.0\text{E}14$. The bottom drag coefficient and the vertical viscosity are set to $c_{\text{drag}} = 1.0\text{E}-3$ and $\nu_v = 1.0\text{E}-4$, respectively.

For the convergence test, we consider a short time interval with a terminal time of $T = 4096$ s, and we use Eq. (3.3) to advance the baroclinic mode in the IMEX2 scheme. We compute the relative l_2 errors of the velocity and layer thickness approximations at the terminal time compared to the reference solution generated with $\Delta t = 0.25$ seconds. The numerical results are presented in Table 3, indicating second-order accuracy in time. Next, to demonstrate the performance of the proposed IMEX2 scheme, we consider a

Table 3: Numerical results on relative l_2 errors and convergence rates of u_1 and h_1 at the terminal time $T=4096$ s for the global ocean test case produced by IMEX2 scheme.

Δt	Velocity		Layer thickness	
	Error	Rate	Error	Rate
512	5.5510E-04	-	4.9009E-07	-
256	1.3979E-04	1.99	1.2713E-07	1.95
128	3.4960E-05	2.00	3.2082E-08	1.99
64	8.7483E-06	2.00	8.0385E-09	2.00
32	2.2152E-06	1.98	2.0103E-09	2.00

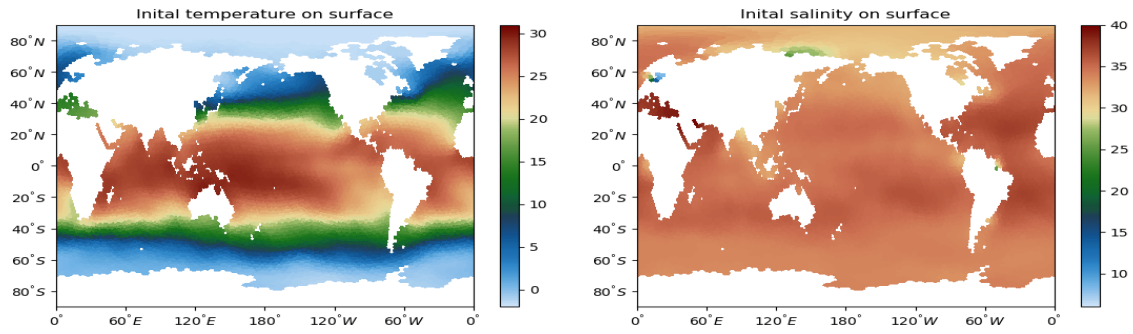


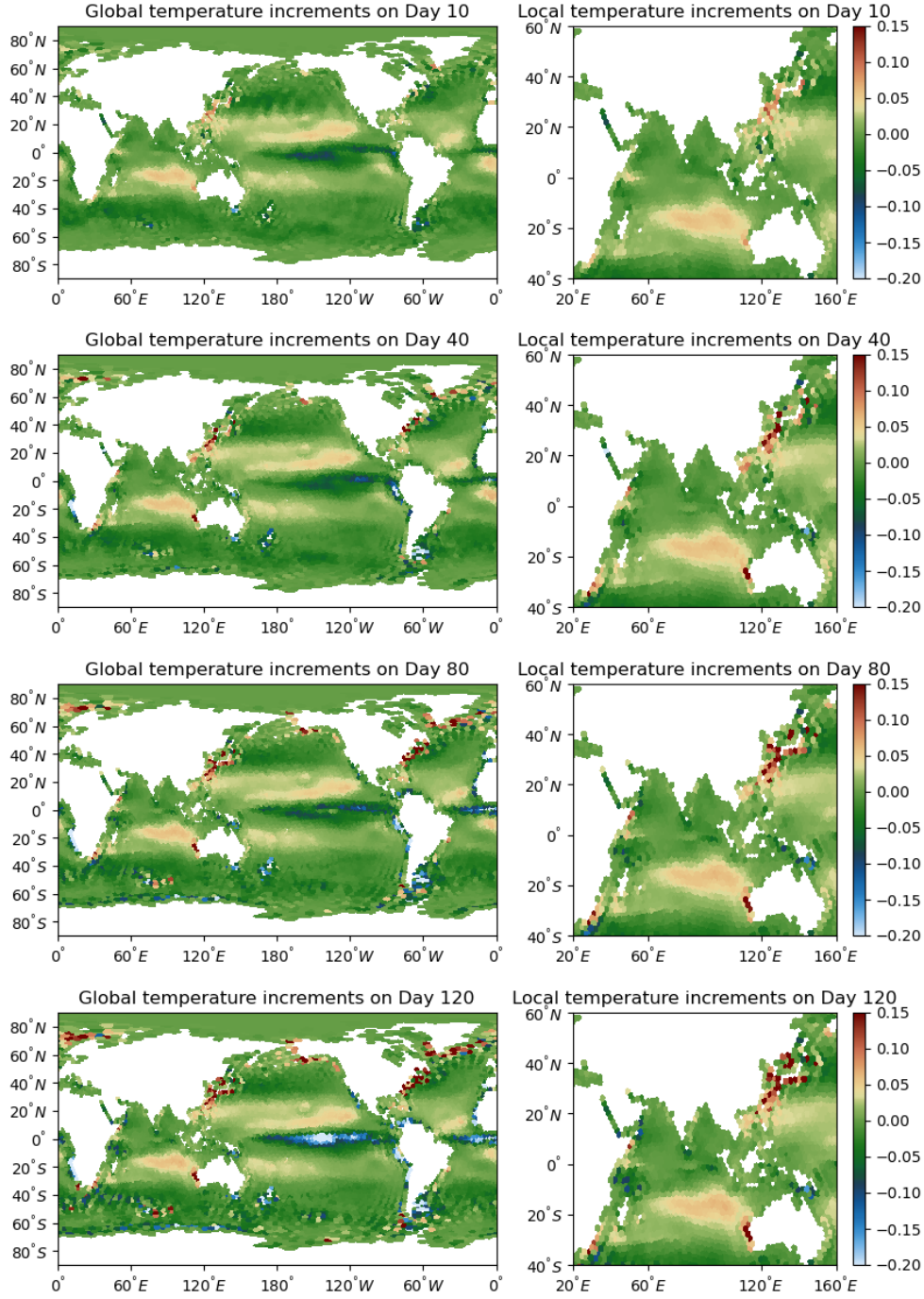
Figure 2: The initial surface temperature and salinity for the global ocean test case.

long-term simulation. To allow for the use of large time step sizes, we adopt the vertical mixing technique, i.e., Eq. (3.5) (“BaroclinicFEulerMixing”), in the IMEX2 scheme, as suggested in MPAS-Ocean. We carry out a 120-day-long simulation for the global ocean test case, using the IMEX2 scheme with $\Delta t = 1800$ s. The initial temperature and salinity values at the ocean surface are presented in Fig. 2, and the simulated surface temperature and salinity increments on Day 10, 40, 80, and 120 are shown in Figs. 3 and 4. The IMEX2 scheme proves to be stable for this long-term simulation, and we achieve similar results as those reported in [13].

5.3 Parallel performance

In this subsection, we evaluate the parallel efficiency of the proposed IMEX2 scheme on a more refined ocean mesh, referred to as “EC60to30,” which is available on the MPAS-Ocean platform. The mesh consists of 60 vertical layers with a variable horizontal resolution ranging from 30 km at the equator and poles to 60 km at the mid-latitudes. Each layer of the mesh includes 235,160 cells, 714,274 edges, and 478,835 vertices.

The domain decomposition is performed solely along the horizontal directions following the routine of the MPAS-Ocean framework and examples of this decomposition



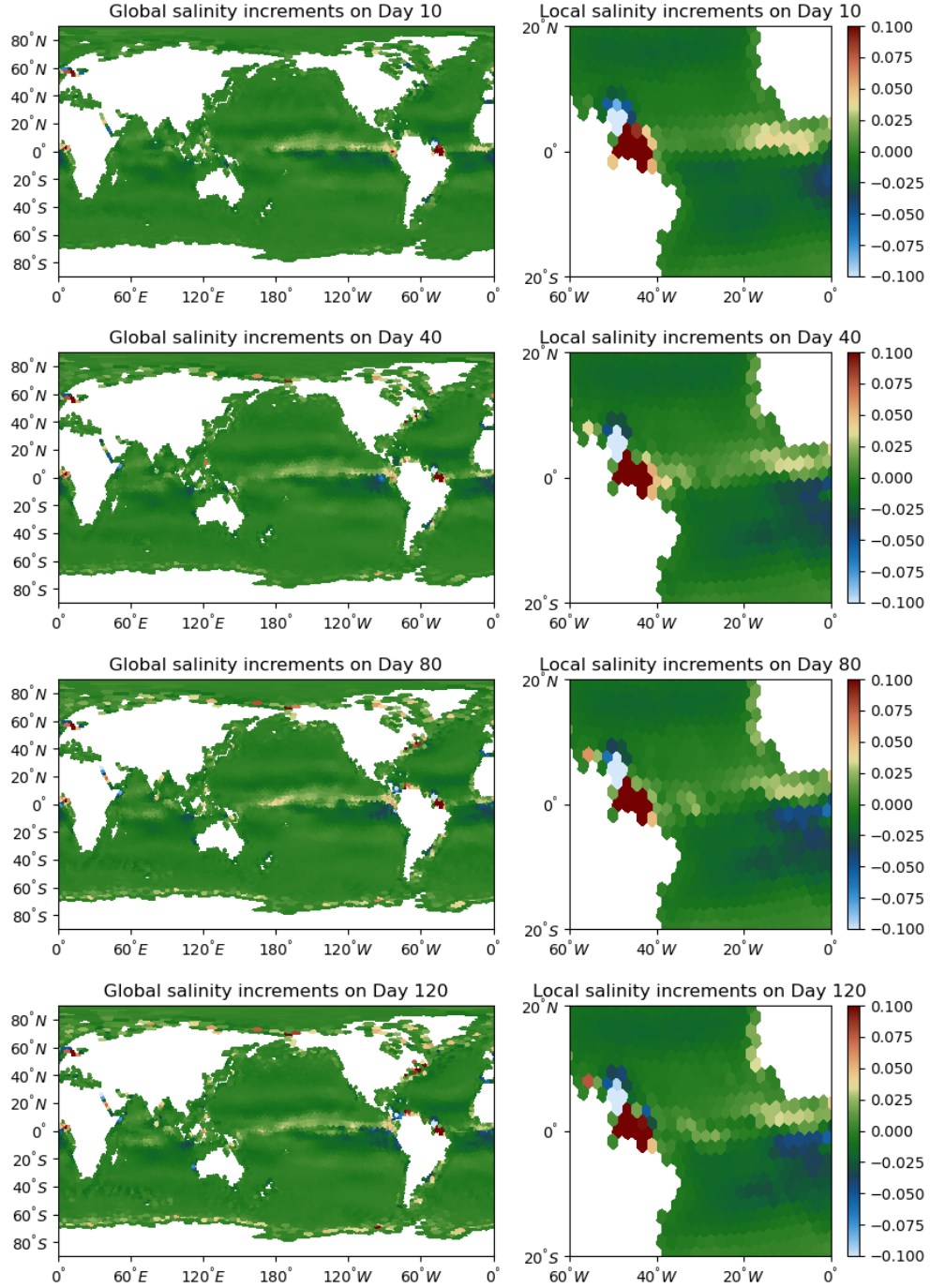


Figure 4: Simulated surface salinity increments for the global ocean test case using the IMEX2 scheme (with vertical mixing) with a time step of $\Delta t = 1800$ s. Results for Day 10, 40, 80, and 120 are shown from top to bottom. The left-hand side figures present the global salinity increments, whereas the right-hand side figures depict the local area between latitudes 20°S and 20°N and longitudes 60°W to 0°E .

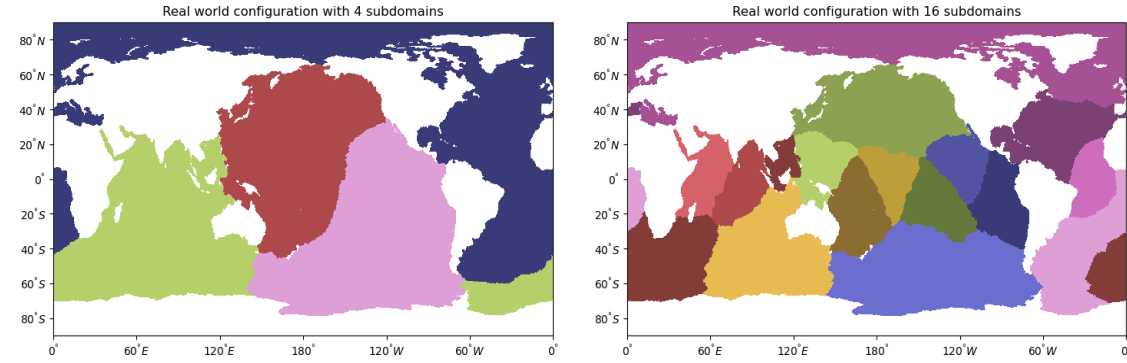


Figure 5: Sample domain decompositions (4 and 16 subdomains) for the global ocean test case generated by the MPAS-Ocean platform. Note that the white part is the land, and the other colored domains are the ocean.

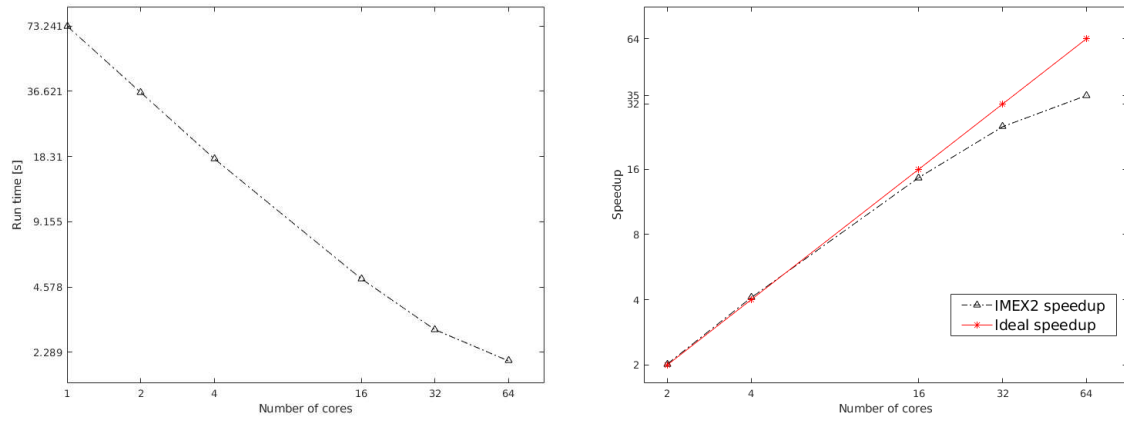


Figure 6: Plots of the running times per time step vs. the number of cores (left) and the corresponding parallel speedups (right) for the global ocean test case.

are shown in Fig. 5. As a result, the unknowns from all layers at the same vertical line stay in the same computing core. To test the parallel performance, we fix the time step size $\Delta t = 600$ s and the terminal time $T = 6000$ s and run the simulations with up to 64 cores. The plots of the running time per time step vs. the number of cores and the corresponding speedup, presented in Fig. 6, show that the IMEX2 scheme achieves good scalability for all tested numbers of cores and produces a speedup of 35 with 64 cores. Moreover, the speedups are very close to the ideal speedups for 2 to 32 cores. Compared to the SE solvers [13,41], the scalability of IMEX2 is relatively worse as expected. There are several potential reasons for the scalability drop when 64 cores are used. As an implicit solver, the CN scheme (part of the IMEX2) needs to form the coefficient matrix and then perform the matrix-vector multiplications for its iterative solver, all of which require intensive inter-processor communications. In addition, each processor may not

be fed enough computing workload in the case of 64 cores (thus produces a lower ratio of computing vs. communication) considering the tested problem size.

6 Conclusions

In this paper, we have proposed a second-order IMEX2 scheme for time-stepping the baroclinic-barotropic split system of primitive equations in layered ocean models. At each time step, we applied the SSPRK2 scheme for explicit advancing of the 3D baroclinic mode and layer thickness, and the linearized CN scheme for implicit solving of the 2D barotropic mode through a coupling process with specially corrected barotropic forcing. We also reconciled the sea-surface height perturbation via flux-form reconciliation. We performed numerical tests on two benchmark test problems from the MPAS-Ocean platform to demonstrate the accuracy, stability, and scalability of the proposed IMEX2 scheme. However, there are still some important questions and tasks worthy of further study. A thorough convergence analysis of the proposed IMEX2 scheme in fully discrete settings, including time and space, is challenging since errors from time integration, spatial discretization, mode splitting, and vertical mixing are intertwined. Developing appropriate vertical mixing techniques that do not affect the overall accuracy when combined with the proposed IMEX2 scheme and other high-order SE schemes would be highly useful in practice. Finally, designing efficient and scalable preconditioned linear system solvers for the barotropic mode of the split system remains an important task for future research, and the elegant work done in [5] provides good guidance in this direction.

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